

Experiment 6: Image Segmentation using KMeans (Color and Spatial Features)

Aim:

To segment images considering both colour and spatial location using KMeans clustering.

Theory:

KMeans clustering is an unsupervised learning algorithm used to group similar data points based on their features. In image segmentation, each pixel is treated as a data point and clustering is performed on its features. There are multiple ways to apply KMeans in image segmentation: • Using only color features like RGB or BGR (each pixel is a 3D vector) • Trying different values of clusters (k) to compare over-segmentation and under-segmentation • Combining both color and spatial coordinates (X, Y) to form a 5D feature vector [B, G, R, X, Y], which helps cluster similar-colored regions that are also spatially close

Algorithm:

- Start
- Import required libraries (`cv2`, `numpy`, `matplotlib`, `sklearn`)
- Read the image using `cv2.imread()`
- For Color-only Segmentation:
 - - Convert to RGB if needed
 - - Reshape image into (num_pixels x 3) feature array
 - - Apply KMeans clustering for k=5 or other values
 - - Reshape labels back into image and display segmented result
- For Multi-k Segmentation:
 - - Repeat above steps for multiple k values (e.g., 2, 3, 7, 10)
 - - Display results using matplotlib subplots
- For Color + Spatial Segmentation:
 - - Generate X, Y coordinates for each pixel using meshgrid
 - - Concatenate [B,G,R,X,Y] into a 5D feature array
 - - Apply KMeans clustering
 - - Display segmented image and print cluster center info
- End

Flowchart:

Start → Import Libraries → Load Image → [Color-only: Reshape to RGB] OR [Color+XY: Reshape to RGB + Meshgrid X,Y] → Apply KMeans → Reshape labels to image → Display/Print segmented results → End

Viva Questions:

- Why is (X, Y) added to color features for segmentation?
- How does increasing the number of clusters affect the output?
- What does each cluster represent in image segmentation using KMeans?